IT5052 – DATA COMMUNICATION & NETWORKING

COMPREHENSIVE MCQ QUESTION BANK & SELF-ASSESSMENT COMPANION

Units IV (Network Layer Protocols), Unit V (Transport Layer Protocols) & Unit VI (Application Layer)

Total Questions: 123 MCQs **Target Exam:** Internal Examination - 2 **Format:** 4 Options + Verified Key

UNIT IV: NETWORK LAYER PROTOCOLS

Q1. Which layer of the OSI model is the Network Layer? [1. Network Layer Introduction]

A) Layer 3 **B)** Layer 2

C) Layer 4 **D)** Layer 7

✓ **Answer: Layer 3**

Q2. What is the primary protocol data unit (PDU) at the Network Layer? [1. Network Layer Introduction]

A) Packet **B)** Frame

C) Segment **D)** Bit

✓ **Answer: Packet**

Q3. Which networking hardware device primarily operates at the Network Layer (Layer 3)? [1. Network Layer Introduction]

A) Router **B)** Repeater

C) Switch (Layer 2) **D)** Hub

✓ **Answer: Router**

Q4. What is the primary responsibility of the Network Layer in computer networking? [1. Network Layer Introduction]

A) Logical addressing, routing, and packet forwarding **B)** Physical synchronization of raw bits

C) Process-to-process communication **D)** Displaying web pages to users

✓ **Answer: Logical addressing, routing, and packet forwarding**

Q5. Which protocol serves as the core and primary protocol of the Network Layer? [1. Network Layer Introduction]

A) IP (Internet Protocol) **B)** TCP

C) HTTP **D)** Ethernet

✓ **Answer: IP (Internet Protocol)**

Q6. Which of the following characteristics best describes the Internet Protocol (IP)? [2. IP - Internet Protocol]

A) Connectionless and best-effort

B) Connection-oriented and reliable

C) Guaranteed ordered delivery

D) Error-free stream delivery

✓ **Answer: Connectionless and best-effort**

Q7. What does 'best-effort delivery' mean in the context of the Internet Protocol (IP)? [2. IP - Internet Protocol]

A) IP does not guarantee delivery, absence of loss, or packet order

B) Packets are delivered only if retransmissions are paid for

C) Packets are encrypted end-to-end automatically

D) Packets always arrive within 1 millisecond

✓ **Answer: IP does not guarantee delivery, absence of loss, or packet order**

Q8. What is the size of an IPv4 address in bits and bytes? [2. IP - Internet Protocol]

A) 32 bits (4 bytes)

B) 64 bits (8 bytes)

C) 128 bits (16 bytes)

D) 16 bits (2 bytes)

✓ **Answer: 32 bits (4 bytes)**

Q9. What is the size of an IPv6 address in bits and bytes? [2. IP - Internet Protocol]

A) 128 bits (16 bytes)

B) 32 bits (4 bytes)

C) 64 bits (8 bytes)

D) 256 bits (32 bytes)

✓ **Answer: 128 bits (16 bytes)**

Q10. Which protocol layer or protocol can be utilized above IP to provide reliability and ordered data delivery? [2. IP - Internet Protocol]

A) TCP at the Transport Layer

B) Ethernet at Data Link Layer

C) UDP at Transport Layer

D) Physical Layer signaling

✓ **Answer: TCP at the Transport Layer**

Q11. What does the acronym ICMP stand for? [3. ICMPv4]

A) Internet Control Message Protocol

B) Internal Computer Monitoring Protocol

C) Inter-network Channel Management Protocol

D) Internet Connection Message Packet

✓ **Answer: Internet Control Message Protocol**

Q12. What is the primary function of ICMPv4 in computer networks? [3. ICMPv4]

A) Sending error reporting messages and diagnostic information

B) Transferring high-resolution multimedia files

C) Assigning permanent MAC addresses to network interfaces

D) Encrypting application payloads

✓ **Answer: Sending error reporting messages and diagnostic information**

Q13. Which popular network diagnostic utility utilizes ICMP Echo Request and Echo Reply messages? [3. ICMPv4]

A) ping
B) telnet
C) ssh
D) netstat

✓ **Answer: ping**

Q14. How are ICMP messages encapsulated and transmitted over networks? [3. ICMPv4]

A) Inside IP datagram packets
B) Directly in physical layer signals
C) Inside TCP segments
D) Inside HTTP responses

✓ **Answer: Inside IP datagram packets**

Q15. Which of the following represents standard error reporting messages in ICMPv4? [3. ICMPv4]

A) Destination Unreachable and Time Exceeded
B) SYN Flood and Reset
C) ACK Timeout and FIN
D) File Not Found and Bad Gateway

✓ **Answer: Destination Unreachable and Time Exceeded**

Q16. What is the fundamental definition of unicast communication? [4. Unicast Routing - Introduction]

A) One sender transmitting data to one specific receiver (One-to-One)
B) One sender transmitting data to all devices on the network
C) One sender transmitting to a selected group
D) Many senders broadcasting simultaneously

✓ **Answer: One sender transmitting data to one specific receiver (One-to-One)**

Q17. What is the primary role of routing in computer networking? [4. Unicast Routing - Introduction]

A) Selecting the optimal path for packets across networks
B) Converting digital signals to analog voltages
C) Encrypting database transactions
D) Allocating memory to operating system processes

✓ **Answer: Selecting the optimal path for packets across networks**

Q18. When a laptop sends an HTTP web request to a specific web server, which communication model is being used? [4. Unicast Routing - Introduction]

A) Unicast
B) Multicast
C) Broadcast
D) Anycast

✓ **Answer: Unicast**

Q19. Which type of routing algorithm shares distance and next-hop vector information exclusively with neighboring routers? [5. Routing Algorithms]

A) Distance Vector routing
B) Link State routing
C) Flooding algorithm
D) Random walk routing

✓ **Answer: Distance Vector routing**

Q20. Which mathematical algorithm is typically used by Distance Vector routing protocols? [5. Routing Algorithms]

A) Bellman-Ford algorithm

B) Dijkstra algorithm

C) Kruskal algorithm

D) Prim algorithm

✓ **Answer: Bellman-Ford algorithm**

Q21. Which type of routing algorithm builds a complete network topology map and calculates shortest paths? [5. Routing Algorithms]

A) Link State routing

B) Distance Vector routing

C) Hop-count vector routing

D) Static default routing

✓ **Answer: Link State routing**

Q22. Which shortest-path algorithm is utilized by Link State routing protocols? [5. Routing Algorithms]

A) Dijkstra algorithm

B) Bellman-Ford algorithm

C) Floyd-Warshall algorithm

D) Binary search algorithm

✓ **Answer: Dijkstra algorithm**

Q23. Which of the following problems is famously associated with Distance Vector routing algorithms? [5. Routing Algorithms]

A) Slow convergence and routing loops (count-to-infinity)

B) Excessive memory consumption due to full topology maps

C) Inability to work with small networks

D) Mandatory requirement of TCP transport

✓ **Answer: Slow convergence and routing loops (count-to-infinity)**

Q24. What does the acronym RIP stand for in computer networking? [6. Unicast Routing Protocols]

A) Routing Information Protocol

B) Remote Internet Protocol

C) Router Interface Program

D) Resource Identification Protocol

✓ **Answer: Routing Information Protocol**

Q25. What routing metric is used by RIP to determine the best path? [6. Unicast Routing Protocols]

A) Hop count

B) Bandwidth and delay

C) Link reliability and MTU

D) Financial link cost

✓ **Answer: Hop count**

Q26. In RIP, what is the maximum number of usable hops, and what does a hop count of 16 indicate? [6. Unicast Routing Protocols]

A) 15 usable hops; 16 indicates unreachable

B) 16 usable hops; 17 indicates unreachable

C) 30 usable hops; 31 indicates unreachable

D) 255 usable hops; 256 indicates unreachable

✓ **Answer: 15 usable hops; 16 indicates unreachable**

Q27. Which transport protocol and port number does RIP utilize for exchanging routing updates? [6. Unicast Routing Protocols]

A) UDP port 520

B) TCP port 179

C) UDP port 69

D) TCP port 80

✓ **Answer: UDP port 520**

Q28. What does the acronym OSPF stand for? [6. Unicast Routing Protocols]

A) Open Shortest Path First

B) Optimal System Packet Forwarding

C) Open Standard Packet Filter

D) Ordered Shortest Path Function

✓ **Answer: Open Shortest Path First**

Q29. Which type of routing algorithm is OSPF based upon? [6. Unicast Routing Protocols]

A) Link State algorithm (Dijkstra)

B) Distance Vector algorithm (Bellman-Ford)

C) Path Vector algorithm

D) Bellman-Moore algorithm

✓ **Answer: Link State algorithm (Dijkstra)**

Q30. How are OSPF packets transported directly over IP? [6. Unicast Routing Protocols]

A) Directly over IP using protocol number 89 (no TCP/UDP header)

B) Over TCP port 89

C) Over UDP port 89

D) Over ICMP messages

✓ **Answer: Directly over IP using protocol number 89 (no TCP/UDP header)**

Q31. What does the acronym BGP stand for, and what is its primary role? [6. Unicast Routing Protocols]

A) Border Gateway Protocol; exchanges routing information between Autonomous Systems

B) Binary Gateway Protocol; routes internal LAN traffic

C) Broadband Group Protocol; distributes broadcast frames

D) Base Grid Protocol; manages private subnets

✓ **Answer: Border Gateway Protocol; exchanges routing information between Autonomous Systems**

Q32. Which transport layer protocol and destination port does BGP use to ensure reliable route exchanges? [6. Unicast Routing Protocols]

A) TCP port 179

B) UDP port 520

C) TCP port 89

D) UDP port 179

✓ **Answer: TCP port 179**

Q33. What is the definition of multicast communication? [7. Multicast Routing & Basics]

A) Transmitting data from one sender to a selected group of receivers (One-to-many selected group)

B) Transmitting data to every device on the physical network

C) Transmitting data from one sender to one receiver

D) Transmitting data from many senders to one receiver

✓ **Answer: Transmitting data from one sender to a selected group of receivers (One-to-many selected group)**

Q34. What is a primary efficiency advantage of multicasting over sending multiple unicast packets? [7. Multicast Routing & Basics]

A) It conserves network bandwidth by duplicating packets only where paths diverge

B) It guarantees zero latency across the Internet

C) It replaces the need for transport layer ports

D) It eliminates router CPU usage entirely

✓ **Answer: It conserves network bandwidth by duplicating packets only where paths diverge**

Q35. Which IPv4 address class is specifically reserved for multicast addressing? [7. Multicast Routing & Basics]

A) Class D

B) Class A

C) Class B

D) Class E

✓ **Answer: Class D**

Q36. What is the defined IPv4 multicast address range? [7. Multicast Routing & Basics]

A) 224.0.0.0 to 239.255.255.255

B) 192.168.0.0 to 192.168.255.255

C) 240.0.0.0 to 255.255.255.254

D) 10.0.0.0 to 10.255.255.255

✓ **Answer: 224.0.0.0 to 239.255.255.255**

Q37. Which protocol is used by IPv4 hosts and local routers to manage multicast group membership? [7. Multicast Routing & Basics]

A) IGMP (Internet Group Management Protocol)

B) ICMP

C) ARP

D) DHCP

✓ **Answer: IGMP (Internet Group Management Protocol)**

Q38. In IPv6, how are all multicast addresses formatted at their start? [7. Multicast Routing & Basics]

A) They begin with FF (e.g. ff00::/8)

B) They begin with FE80

C) They begin with 2001

D) They begin with FC00

✓ **Answer: They begin with FF (e.g. ff00::/8)**

Q39. How is an IPv6 address written and formatted? [9. IPv6 Addressing]

A) 8 groups of hexadecimal digits separated by colons (:)

B) 4 groups of decimal numbers separated by dots (.)

C) 6 groups of octal digits separated by hyphens (-)

D) 16 binary octets separated by commas

✓ **Answer: 8 groups of hexadecimal digits separated by colons (:)**

Q40. Which compression rule is valid when abbreviating an IPv6 address? [9. IPv6 Addressing]

A) Leading zeros in any group can be omitted, and consecutive zero groups can be replaced with :: once

B) Any zero anywhere can be permanently erased without limit

C) Colons can be omitted if letters are present

D) Double colons (::) can be used multiple times in the same address

✓ **Answer: Leading zeros in any group can be omitted, and consecutive zero groups can be replaced with :: once**

Q41. Which transmission mode exists in IPv4 but was completely eliminated in IPv6? [9. IPv6 Addressing]

A) Broadcast

B) Multicast

C) Unicast

D) Anycast

✓ **Answer: Broadcast**

Q42. Which of the following communication types are supported by IPv6? [9. IPv6 Addressing]

A) Unicast, Multicast, and Anycast

B) Unicast, Broadcast, and Simulcast

C) Broadcast and Multicast only

D) Unicast and Broadcast only

✓ **Answer: Unicast, Multicast, and Anycast**

Q43. What is the fixed size of the IPv6 base header? [10. IPv6 Protocol]

A) 40 bytes

B) 20 bytes

C) 32 bytes

D) 64 bytes

✓ **Answer: 40 bytes**

Q44. What is the minimum base header size of an IPv4 packet (without optional fields)? [10. IPv6 Protocol]

A) 20 bytes

B) 40 bytes

C) 8 bytes

D) 16 bytes

✓ **Answer: 20 bytes**

Q45. Why was the header checksum field removed from the IPv6 base header? [10. IPv6 Protocol]

A) To speed up packet processing at intermediate routers since upper/lower layers provide error checks

B) Because IPv6 packets are immune to physical transmission noise

C) Because IPv6 does not allow routers to forward packets

D) To force routers to calculate CRC-32 on payloads

✓ **Answer: To speed up packet processing at intermediate routers since upper/lower layers provide error checks**

Q46. Which protocol does IPv6 utilize instead of the traditional IPv4 ARP (Address Resolution Protocol)? [10. IPv6 Protocol]

A) Neighbor Discovery Protocol (NDP)

B) RARP

C) BOOTP

D) IGMPv3

✓ **Answer: Neighbor Discovery Protocol (NDP)**

Q47. How does packet fragmentation work in IPv6 compared to IPv4? [10. IPv6 Protocol]

A) Routers do not fragment IPv6 packets; fragmentation is performed only by the sending source device

B) Intermediate routers fragment IPv6 packets whenever MTU is exceeded

C) IPv6 packets can never be fragmented under any circumstance

D) The destination host performs fragmentation prior to acknowledgment

✓ **Answer: Routers do not fragment IPv6 packets; fragmentation is performed only by the sending source device**

Q48. Which control and diagnostics protocol accompanies IPv6? [10. IPv6 Protocol]

A) ICMPv6

B) ICMPv4

C) IGMP

D) SNMPv1

✓ **Answer: ICMPv6**

UNIT V: TRANSPORT LAYER PROTOCOLS

Q49. Which layer of the OSI reference model is the Transport Layer? [1. Transport Layer Introduction]

A) Layer 4

B) Layer 3

C) Layer 5

D) Layer 7

✓ **Answer: Layer 4**

Q50. What fundamental type of communication is provided by the Transport Layer? [1. Transport Layer Introduction]

A) Process-to-process communication between applications on different devices

B) Node-to-node hop transmission between adjacent interfaces

C) Bit synchronization across physical cables

D) Host-to-host network packet routing

✓ **Answer: Process-to-process communication between applications on different devices**

Q51. What is the general name of the Protocol Data Unit (PDU) at the Transport Layer? [1. Transport Layer Introduction]

A) Segment (for TCP) or User Datagram (for UDP)

B) Frame

C) Packet

D) Bit stream

✓ **Answer: Segment (for TCP) or User Datagram (for UDP)**

Q52. What identifier is used at the Transport Layer to distinguish between multiple applications running on the same host? [2. Transport Layer Services]

A) Port number

B) IP address

C) MAC address

D) URL path

✓ **Answer: Port number**

Q53. What is the difference between an IP address and a Port number? [2. Transport Layer Services]

A) IP address identifies a host/interface on a network; Port number identifies a specific application/process on that host

B) IP address identifies an application; Port number identifies the router

C) Both are interchangeable 32-bit identifiers

D) Port numbers are assigned by ISPs, whereas IP addresses are embedded in hardware

✓ **Answer: IP address identifies a host/interface on a network; Port number identifies a specific application/process on that host**

Q54. Which of the following is a standard well-known port example? [2. Transport Layer Services]

A) Port 443 for HTTPS web traffic

B) Port 9999 for IPv6 routing

C) Port 25 for DNS queries

D) Port 80 for SSH remote login

✓ **Answer: Port 443 for HTTPS web traffic**

Q55. What does the combination of an IP address and a Port number form? [2. Transport Layer Services]

A) A Socket address

B) A MAC frame

C) A Subnet mask

D) A DNS record

✓ **Answer: A Socket address**

Q56. What are the three primary transport layer protocols in the modern Internet protocol suite? [3. Transport Layer Protocols Overview]

A) UDP, TCP, and SCTP

B) IP, ICMP, and IGMP

C) HTTP, FTP, and SMTP

D) Ethernet, Wi-Fi, and Token Ring

✓ **Answer: UDP, TCP, and SCTP**

Q57. Which protocol is connectionless and lightweight with minimal overhead? [3. Transport Layer Protocols Overview]

A) UDP (User Datagram Protocol)

B) TCP (Transmission Control Protocol)

C) SCTP

D) BGP

✓ **Answer: UDP (User Datagram Protocol)**

Q58. Which protocol provides reliable, connection-oriented byte stream transmission? [3. Transport Layer Protocols Overview]

A) TCP (Transmission Control Protocol)

B) UDP

C) IP

D) ICMP

✓ **Answer: TCP (Transmission Control Protocol)**

Q59. Which protocol is reliable and message-oriented, supporting multi-homing and multi-streaming? [3. Transport Layer Protocols Overview]

A) SCTP (Stream Control Transmission Protocol)

B) UDP

C) TCP

D) OSPF

✓ **Answer: SCTP (Stream Control Transmission Protocol)**

Q60. What does the acronym UDP stand for? [4. UDP]

A) User Datagram Protocol

B) Unified Data Packet

C) Universal Delivery Protocol

D) Unchecked Datagram Program

✓ **Answer: User Datagram Protocol**

Q61. What is the size of the standard UDP header? [4. UDP]

A) 8 bytes

B) 20 bytes

C) 40 bytes

D) 16 bytes

✓ **Answer: 8 bytes**

Q62. Which fields constitute the 8-byte UDP header? [4. UDP]

A) Source Port, Destination Port, Length, and Checksum

B) Sequence Number, Acknowledgment Number, Window Size, and Flags

C) Version, TTL, Protocol, and Checksum

D) IP Source, IP Destination, MAC Source, and MAC Destination

✓ **Answer: Source Port, Destination Port, Length, and Checksum**

Q63. Why is UDP chosen for applications like online gaming, video conferencing, and DNS queries? [4. UDP]

A) It has very low overhead and introduces no connection setup or retransmission delays

B) It encrypts packets automatically with zero CPU overhead

C) It guarantees 100% error-free delivery

D) It reserves dedicated optical bandwidth on routers

✓ **Answer: It has very low overhead and introduces no connection setup or retransmission delays**

Q64. Does UDP provide guaranteed packet order or retransmission of lost datagrams? [4. UDP]

A) No, UDP itself provides neither delivery guarantee nor retransmission

B) Yes, UDP automatically retransmits missing datagrams three times

C) Yes, UDP reorders packets using sequence numbers

D) Only when running on Gigabit Ethernet

✓ **Answer: No, UDP itself provides neither delivery guarantee nor retransmission**

Q65. What does the acronym TCP stand for? [5. TCP]

A) Transmission Control Protocol

B) Transfer Communication Packet

C) Terminal Connection Protocol

D) Transport Control Packet

✓ **Answer: Transmission Control Protocol**

Q66. What is the minimum header size of a TCP segment (without options)? [5. TCP]

A) 20 bytes

B) 8 bytes

C) 40 bytes

D) 16 bytes

✓ **Answer: 20 bytes**

Q67. How does TCP achieve reliable and ordered delivery of application data? [5. TCP]

A) By using sequence numbers, positive acknowledgments, and retransmission timers

B) By broadcasting all segments to every connected device

C) By preventing packet loss at the physical layer

D) By relying on the underlying IP layer for guaranteed delivery

✓ **Answer: By using sequence numbers, positive acknowledgments, and retransmission timers**

Q68. What is the sequence of steps in the standard TCP Three-Way Handshake for connection establishment? [5. TCP]

A) SYN -> SYN-ACK -> ACK

B) ACK -> SYN -> SYN-ACK

C) SYN -> ACK -> DATA

D) HELLO -> ACK -> CONNECT

✓ **Answer: SYN -> SYN-ACK -> ACK**

Q69. What occurs in step 1 of the TCP three-way handshake? [5. TCP]

A) The client sends a SYN packet to request connection synchronization

B) The server responds with an ACK

C) The client sends payload data immediately

D) The connection is abruptly terminated

✓ **Answer: The client sends a SYN packet to request connection synchronization**

Q70. What occurs in step 2 of the TCP three-way handshake? [5. TCP]

A) The server accepts the connection request and responds with a SYN-ACK packet

B) The client sends an ACK acknowledgment

C) The server closes the socket immediately

D) The client and server exchange symmetric encryption keys

✓ **Answer: The server accepts the connection request and responds with a SYN-ACK packet**

Q71. What occurs in step 3 of the TCP three-way handshake? [5. TCP]

A) The client confirms receipt with an ACK packet, completing connection setup

B) The server resets the connection with RST

C) The client sends a FIN request

D) The server retransmits the SYN packet

✓ **Answer: The client confirms receipt with an ACK packet, completing connection setup**

Q72. Which control mechanisms are natively built into TCP to manage network transmission speeds? [5. TCP]

A) Flow control (Sliding Window) and Congestion Control

B) Frequency division multiplexing

C) Spread spectrum modulation

D) Hardware interrupt arbitration

✓ **Answer: Flow control (Sliding Window) and Congestion Control**

Q73. Which flag in the TCP header is used to gracefully close/terminate an established connection? [5. TCP]

A) FIN

B) SYN

C) URG

D) PSH

✓ **Answer: FIN**

Q74. What does the acronym SCTP stand for? [6. SCTP]

A) Stream Control Transmission Protocol

B) Synchronous Communication Transfer Protocol

C) Socket Connection Transport Protocol

D) Simple Control Transmission Packet

✓ **Answer: Stream Control Transmission Protocol**

Q75. Which feature allows SCTP to handle multiple independent message streams within a single connection? [6. SCTP]

A) Multi-streaming
B) Multi-homing
C) Subnetting
D) Frequency hopping

✓ **Answer: Multi-streaming**

Q76. Which feature in SCTP allows an endpoint to bind to multiple IP addresses for link redundancy and failover? [6. SCTP]

A) Multi-homing
B) Multi-streaming
C) Broadcasting
D) Sliding window

✓ **Answer: Multi-homing**

Q77. What term does SCTP use to refer to an established connection between two endpoints? [6. SCTP]

A) Association
B) Socket pair
C) Virtual circuit
D) Pipe

✓ **Answer: Association**

Q78. How does SCTP transmit data in terms of framing? [6. SCTP]

A) Message-oriented (preserves message boundaries)
B) Continuous unformatted byte stream
C) Raw bit frames only
D) Encrypted blocks only

✓ **Answer: Message-oriented (preserves message boundaries)**

UNIT VI: APPLICATION LAYER

Q79. Which layer of the OSI model is the Application Layer? [1. Application Layer Introduction]

A) Layer 7 (highest layer)
B) Layer 1
C) Layer 4
D) Layer 5

✓ **Answer: Layer 7 (highest layer)**

Q80. What is the main role of the Application Layer in computer networks? [1. Application Layer Introduction]

A) To provide network services and interfaces directly to user applications
B) To route packets through intermediate transit routers
C) To manage physical voltage levels on copper wires
D) To regulate node-to-node MAC addresses

✓ **Answer: To provide network services and interfaces directly to user applications**

Q81. Which of the following is a collection of common Application Layer protocols? [1. Application Layer Introduction]

A) HTTP, FTP, SMTP, DNS, and SSH

B) IP, ICMP, ARP, and RARP

C) TCP, UDP, and SCTP

D) Ethernet, PPP, and HDLC

✓ **Answer: HTTP, FTP, SMTP, DNS, and SSH**

Q82. In the Client-Server networking architecture, which role does the client play? [2. Client-Server Programming]

A) The client initiates communication by sending a request for a service

B) The client waits passively and responds to server requests

C) The client controls the physical network routing tables

D) The client assigns MAC addresses to all connecting machines

✓ **Answer: The client initiates communication by sending a request for a service**

Q83. In the Client-Server model, which role does the server play? [2. Client-Server Programming]

A) The server continuously listens for requests, processes them, and returns responses

B) The server initiates connections to random internet clients

C) The server fragments physical layer frames

D) The server terminates local computer power

✓ **Answer: The server continuously listens for requests, processes them, and returns responses**

Q84. Can a single server simultaneously communicate with and serve multiple clients? [2. Client-Server Programming]

A) Yes, a server can handle multiple concurrent client connections

B) No, a server can only communicate with exactly one client at a time

C) Only if all clients share the same IP address

D) Only if client and server run the identical operating system version

✓ **Answer: Yes, a server can handle multiple concurrent client connections**

Q85. Are the Internet and the World Wide Web (WWW) the same thing? [3. WWW and HTTP]

A) No: the Internet is the underlying network infrastructure; the WWW is a service/collection of resources accessed over the Internet

B) Yes: both terms are 100% identical and interchangeable technical terms

C) No: the WWW is physical cabling, while the Internet is software only

D) Yes: the Internet was renamed to WWW in 1999

✓ **Answer: No: the Internet is the underlying network infrastructure; the WWW is a service/collection of resources accessed over the Internet**

Q86. What does the acronym HTTP stand for? [3. WWW and HTTP]

A) Hypertext Transfer Protocol

B) High-speed Text Transmission Protocol

C) Host Terminal Transfer Packet

D) Hyperlink Text Translation Program

✓ **Answer: Hypertext Transfer Protocol**

Q87. Why is HTTP classified as a 'stateless' protocol? [3. WWW and HTTP]

A) Because the server retains no memory of previous client requests across requests

B) Because it cannot run on computers that have hard drives

C) Because it transmits state governments data only

D) Because connection states are stored on DNS root servers

✓ **Answer: Because the server retains no memory of previous client requests across requests**

Q88. What is the standard default port number for unencrypted HTTP traffic? [3. WWW and HTTP]

A) Port 80

B) Port 443

C) Port 21

D) Port 25

✓ **Answer: Port 80**

Q89. What is the standard default port number for secure, encrypted HTTPS traffic? [3. WWW and HTTP]

A) Port 443

B) Port 80

C) Port 22

D) Port 53

✓ **Answer: Port 443**

Q90. What security protocol does HTTPS use to provide encrypted communication between web browser and server? [3. WWW and HTTP]

A) TLS (Transport Layer Security) / SSL

B) IPsec only

C) WPA3

D) WEP

✓ **Answer: TLS (Transport Layer Security) / SSL**

Q91. What does the acronym FTP stand for? [4. FTP]

A) File Transfer Protocol

B) Fast Transmission Packet

C) Folder Tracking Program

D) Format Translation Protocol

✓ **Answer: File Transfer Protocol**

Q92. Which underlying transport protocol does FTP use to guarantee reliable file transfers? [4. FTP]

A) TCP

B) UDP

C) ICMP

D) IPsec

✓ **Answer: TCP**

Q93. FTP famously uses two separate connections. Which port is commonly used for the control connection? [4. FTP]

A) Port 21

B) Port 20

C) Port 80

D) Port 25

✓ **Answer: Port 21**

Q94. In traditional active-mode FTP, which port number is commonly used for the data transfer connection? [4. FTP]

A) Port 20

B) Port 21

C) Port 22

D) Port 110

✓ **Answer: Port 20**

Q95. What is the difference between FTP's control connection and data connection? [4. FTP]

A) Control connection (port 21) sends commands and replies; data connection (port 20) transfers actual file contents

B) Control transfers files; data transfers user passwords

C) Control connection uses UDP; data connection uses TCP

D) There is no difference; both transfer random binary packets

✓ **Answer: Control connection (port 21) sends commands and replies; data connection (port 20) transfers actual file contents**

Q96. What is the primary role of the Simple Mail Transfer Protocol (SMTP)? [5. Electronic Mail]

A) Sending and relaying email messages from client to server or between mail servers

B) Downloading email messages to offline clients

C) Browsing web mail pages

D) Translating domain names into IP addresses

✓ **Answer: Sending and relaying email messages from client to server or between mail servers**

Q97. Which protocol is traditionally used by email clients to retrieve and download emails from a mail server to a local device? [5. Electronic Mail]

A) POP3 (Post Office Protocol version 3)

B) SMTP

C) SNMP

D) SSH

✓ **Answer: POP3 (Post Office Protocol version 3)**

Q98. Which email retrieval protocol allows users to access, organize into folders, and synchronize emails directly on the mail server? [5. Electronic Mail]

A) IMAP (Internet Message Access Protocol)

B) POP3

C) SMTP

D) FTP

✓ **Answer: IMAP (Internet Message Access Protocol)**

Q99. What are the common standard port numbers associated with SMTP? [5. Electronic Mail]

A) Port 25 (traditional), 587 (submission), and 465 (secure)

B) Port 80 and 443

C) Port 20 and 21

D) Port 110 and 995

✓ **Answer: Port 25 (traditional), 587 (submission), and 465 (secure)**

Q100. What are the standard default port numbers for POP3 (unencrypted and secure)? [5. Electronic Mail]

A) Port 110 (standard) and Port 995 (secure SSL/TLS)

B) Port 143 and Port 993

C) Port 25 and Port 465

D) Port 21 and Port 990

✓ **Answer: Port 110 (standard) and Port 995 (secure SSL/TLS)**

Q101. What are the standard default port numbers for IMAP (unencrypted and secure)? [5. Electronic Mail]

A) Port 143 (standard) and Port 993 (secure SSL/TLS)

B) Port 110 and Port 995

C) Port 25 and Port 587

D) Port 80 and Port 443

✓ **Answer: Port 143 (standard) and Port 993 (secure SSL/TLS)**

Q102. What does the acronym SSH stand for? [6. SSH]

A) Secure Shell

B) System Security Host

C) Socket Secure Handler

D) Server Sharing Header

✓ **Answer: Secure Shell**

Q103. What is the primary function of SSH in network administration? [6. SSH]

A) Providing encrypted remote login and command-line execution over insecure networks

B) Sending audio-video broadcasts to multiple computers

C) Assigning DHCP dynamic IP addresses

D) Filtering spam emails on mail servers

✓ **Answer: Providing encrypted remote login and command-line execution over insecure networks**

Q104. What is the default TCP port number used by SSH servers? [6. SSH]

A) Port 22

B) Port 23

C) Port 21

D) Port 25

✓ **Answer: Port 22**

Q105. Which legacy, insecure protocol does SSH commonly replace for remote command-line access? [6. SSH]

A) Telnet (which transmits passwords in plaintext)

B) FTP

C) DNS

D) HTTPS

✓ **Answer: Telnet (which transmits passwords in plaintext)**

Q106. What does the acronym DNS stand for, and what is its main purpose? [7. DNS]

A) Domain Name System; translates human-friendly domain names into IP addresses

B) Data Network Service; compresses video packets

C) Dynamic Node Synchronization; assigns MAC addresses

D) Digital Name Server; encrypts email text

✓ **Answer: Domain Name System; translates human-friendly domain names into IP addresses**

Q107. What is the standard port number used by DNS servers? [7. DNS]

A) Port 53

B) Port 25

C) Port 80

D) Port 110

✓ **Answer: Port 53**

Q108. Which transport layer protocols can DNS utilize? [7. DNS]

A) Both UDP and TCP (UDP for standard queries; TCP for zone transfers and large responses)

B) UDP only

C) TCP only

D) ICMP only

✓ **Answer: Both UDP and TCP (UDP for standard queries; TCP for zone transfers and large responses)**

Q109. Which DNS record type maps a domain name to a 32-bit IPv4 address? [7. DNS]

A) A record

B) AAAA record

C) MX record

D) CNAME record

✓ **Answer: A record**

Q110. Which DNS record type maps a domain name to a 128-bit IPv6 address? [7. DNS]

A) AAAA record

B) A record

C) NS record

D) TXT record

✓ **Answer: AAAA record**

Q111. Which DNS record type specifies the mail server responsible for receiving emails on behalf of a domain? [7. DNS]

A) MX record

B) CNAME record

C) PTR record

D) A record

✓ **Answer: MX record**

Q112. Which DNS record type creates an alias pointing one domain name to another canonical domain name? [7. DNS]

A) CNAME record

B) A record

C) NS record

D) SOA record

✓ **Answer: CNAME record**

Q113. Which DNS record type delegates a DNS zone to use specific authoritative name servers? [7. DNS]

A) NS record

B) A record

C) MX record

D) AAAA record

✓ **Answer: NS record**

Q114. What is the primary objective of data compression in computer networks? [8. Data Compression]

A) To reduce data size to save storage space and transmission time

B) To encrypt plain text to prevent eavesdropping

C) To convert binary data into human-readable characters

D) To assign unique port numbers to applications

✓ **Answer: To reduce data size to save storage space and transmission time**

Q115. What is the defining characteristic of Lossless compression? [8. Data Compression]

A) No information is lost; the original data can be reconstructed perfectly

B) Some information is permanently discarded to maximize size reduction

C) It only compresses audio files

D) Data cannot be decompressed once compressed

✓ **Answer: No information is lost; the original data can be reconstructed perfectly**

Q116. Which of the following are prime examples of Lossless compression formats? [8. Data Compression]

A) ZIP and PNG

B) JPEG and MP3

C) MPEG video and MP3 audio

D) Lossy WebP audio

✓ **Answer: ZIP and PNG**

Q117. What is the defining characteristic of Lossy compression? [8. Data Compression]

A) Some information is permanently removed, achieving higher compression ratios where exact precision is not critical

B) Original data can always be reconstructed bit-for-bit

C) It is used exclusively for text source code and database backups

D) It never reduces file size

✓ **Answer: Some information is permanently removed, achieving higher compression ratios where exact precision is not critical**

Q118. Which of the following are prime examples of Lossy compression formats? [8. Data Compression]

A) JPEG and MP3

B) ZIP and 7z

C) PNG and FLAC

D) TAR and GZ

✓ **Answer: JPEG and MP3**

UNIT IV: NETWORK LAYER PROTOCOLS

Q119. Which comparison between IPv4 and IPv6 is correct regarding address format? [Comparison: IPv4 vs IPv6]

A) IPv4 uses decimal dotted notation; IPv6 uses hexadecimal colon notation

B) IPv4 uses colons; IPv6 uses decimal dots

C) Both use dotted hexadecimal notation

D) IPv4 is 128-bit; IPv6 is 32-bit

✓ **Answer: IPv4 uses decimal dotted notation; IPv6 uses hexadecimal colon notation**

Q120. Which feature is present in the IPv4 base header but omitted in the IPv6 base header to increase router forwarding performance? [Comparison: IPv4 vs IPv6]

A) Header Checksum

B) Hop Limit

C) Source Address

D) Payload Length

✓ **Answer: Header Checksum**

Q121. Which pair correctly matches the routing protocol with its underlying algorithm? [Comparison: Routing Protocols]

A) RIP -> Bellman-Ford; OSPF -> Dijkstra

B) RIP -> Dijkstra; OSPF -> Bellman-Ford

C) BGP -> Dijkstra; RIP -> Prim

D) OSPF -> Bellman-Ford; BGP -> Floyd

✓ **Answer: RIP -> Bellman-Ford; OSPF -> Dijkstra**

UNIT V: TRANSPORT LAYER PROTOCOLS

Q122. Which comparison correctly pairs transport protocol header sizes? [Comparison: TCP vs UDP]

A) TCP minimum header is 20 bytes; UDP header is 8 bytes

B) TCP header is 8 bytes; UDP header is 20 bytes

C) Both headers are exactly 40 bytes

D) TCP header is 4 bytes; UDP header is 16 bytes

✓ **Answer: TCP minimum header is 20 bytes; UDP header is 8 bytes**

UNIT VI: APPLICATION LAYER

Q123. Which statement accurately distinguishes SMTP from POP3 and IMAP? [Comparison: Email Protocols]

A) SMTP is used to push/send email; POP3 and IMAP are used to retrieve/access email

B) SMTP retrieves emails; POP3 sends emails

C) SMTP operates on port 110; POP3 operates on port 25

D) IMAP is connectionless; SMTP is connection-oriented

✓ **Answer: SMTP is used to push/send email; POP3 and IMAP are used to retrieve/access email**